

SATYANARAYANA CHILLALE

Senior Software Engineer | 6+ Years | Full Stack | Cloud | AI/ML

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WORK EXPERIENCE

Oracle

Bangalore, India

Senior Member Technical Staff

May 2021 – Aug 2023

- **Orchestrated distributed data consistency:** by developing and launching a cloud-native Go service – MicroTx
- Scaled the system to process over **300 billion** transactions at **8,500 transactions/sec**, with failure rate below **0.01%**
- Engineered strong data consistency via industry-standard XA protocol supporting MySQL, Postgres, and Oracle DB
- Implemented eventual consistency through LRA and TCC protocols for stateless NoSQL DB
- Designed application-layer middleware APIs to support Java, Node.js, and Python microservices
- Integrated out-of-the-box Istio API Gateway authorization for secure access via OAuth2, JWT, and enterprise SSO
- Built unified Helm charts to enable seamless deployment on AWS, Oracle Cloud and GCP Kubernetes environment
- Developed admin console backend APIs to support real-time transaction monitoring, tracing, and service health checks
- Mentored three junior developers on building the admin console UI for MicroTx
- **Oracle Blockchain Platform on Kubernetes:** Designed the platform's cloud-native architecture as the technical lead
- Adapted Helm deployment patterns to support cloud blockchain provisioning in **10 minutes** across cloud environments
- Extended expertise from MicroTx to configure secure SSL and certificate management across blockchain nodes
- Leveraged Istio and Envoy to manage traffic routing for 10+ blockchain services with SNI-based ingress

SAP Labs

Bangalore, India

Senior Software Developer

April 2019 – May 2021

- **Led the design of Server-driven UI system:** that automated front-end generation and eliminated manual development
- Reduced delivery timelines from **12 weeks to 4 weeks** while saving **40% resources of 75 engineers**
- Defined flexible JSON schema specifications to model versatile and dynamic UI components, layouts and behaviors
- **RESTful Application Model:** Built tools adopted by 10,000+ engineers to accelerate ERP app development
- Below tools boosted overall ERP application **development efficiency by 30%** across service teams
- Auto-generated core entities and all service artifacts from API specifications to consume remote services
- Created tool to generate OData API specifications from SAP core data models, enabling faster API delivery
- Developed one-click REST API and web UI auto-generator, improving development speed and testing

Software Developer

July 2017 – March 2019

- Recognized with **Best Early Talent Award** in **2018** and **2019** for top performance and, high-impact contributions
- Implemented fine-grained access control for Data Preview, enabling secure exploration of 10,000+ S/4 HANA tables
- Enhanced Data Preview tool with advanced filtering and SQL editor capabilities to improve data exploration

TECHNICAL SKILLS

Languages: Java (OOP), Go, C++, Python, TypeScript

Backend: Node.js, REST, GraphQL, Spark, Helidon, Django

Cloud & DevOps: Docker, Kubernetes, Istio, Helm, AWS

Frontend: React, Next.js, Vue.js

Databases: MySQL, PostgreSQL, MongoDB, etc

ML: PyTorch, scikit-learn, Pandas, NumPy, CUDA

PROJECTS

- **Image Geolocation (2025) :** Built an advanced ML model for image geolocation trained on 4 million images using CLIP and RAG techniques. Innovated a novel hierarchical feature clustering inference approach for location search. **Reduced the search space by 100x** while maintaining accuracy comparable to state-of-the-art methods. Achieved competitive accuracy 85.08% (continent level) and 15.01% (street level) on the IM2GPS3k benchmark
- **Future Video Frame Prediction (2024) :** A ML model to predict the future frames in videos containing moving objects. Implemented a UNet-based architecture for mask generation with **96% accuracy**. Trained a Conditional Masked Video for enhanced temporal consistency and accuracy. **Top 5 projects** in the Deep Learning course

EDUCATION

New York University, Courant Institute of Mathematical Sciences

New York, NY

Master of Science in Computer Science; **GPA: 3.9/4.0**

Aug 2023 – May 2025

- **Systems courses:** Distributed Systems, Big Data Application Development, Honors Analysis of Algorithms
- **ML courses:** GPUs, Reinforcement Learning, Deep Learning, Gaps in Modern Machine Learning, Computer Vision

PES University

Bangalore, India

Bachelor of Engineering in Computer Science; **GPA: 3.6/4.0**

Aug 2013 – June 2017